Computer Science Practical List 6
(SQL Assignment to be solved using MySQL)

Before attempting the following questions, ensure that you have created a database and opened it to perform various operations using **SQL**.

Write SQL commands for the following:

1. To create a table TEACHER with the following Columns:
   TCODE VARCHAR(3) PRIMARY KEY, TEACHERNAME VARCHAR(25), STREAM VARCHAR(15), BASIC FLOAT, DATEOFJOINING DATE
2. To insert the following rows in the table TEACHER

   Table: TEACHER

TCODE	TEACHERNAME	STREAM	BASIC	DATEOFJOINING
101	ANANYA MURTY	SCIENCE	18000	1990-01-23
202	KIRTI KHANEJA	HUMANITIES	28000	1978-12-12
103	ADIL MEHRA	SCIENCE	8000	2001-02-14
305	NISHIYA GOEL	COMMERCE	12000	1997-01-01
203	SHUBH PANDIT	HUMANITIES	22000	1985-03-19
109	NAINA	SCIENCE	20000	1994-07-17

3. To create a table STUDENT with the following Columns:

SCODE	**INT**
NAME	**VARCHAR(20)**
TCODE	**VARCHAR(3)**
AGG	**FLOAT**

4. To insert the following rows in the table STUDENT:

   Table: STUDENT

SCODE	NAME	TCODE	AGG
2	NABI AHMAD	101	91
1	RAVI SAHAI	305	84
5	VIBHUTI	203	67
4	NAZNEEN	103	89
3	ARYANA	202	37
6	JONATHAN	305	45

5. To display the complete contents of the table TEACHER
6. To display details of those Teachers whose date of joining is from 1st of Jan 1978 (‘1978-01-01’) to 30th of November 1992 (‘1992-11-30’ - including both the days)
7. To display the contents of the table STUDENT in descending order of AGG.
8. To display NAME and SCODE of all the students who are taught by teachers having code numbers 101 and 203
9. To display income tax amount from the table TEACHER by computing income tax as 25% of BASIC
10. To replace the value of column AGG with 82 where NAME is Ravi Sahai in the table STUDENT
11. To delete all the records from the table STUDENT where AGG is below 40.
12. To change the STREAM of the teacher whose name is Ananya Murty to 'Humanities' and her TCODE to 210.
13. To add a column called GRADE in the table STUDENT to accommodate one character.
14. To replace the column GRADE with 'A' in the table STUDENT whose AGG>=70
15. To replace the column GRADE with 'B' in the table STUDENT whose AGG>=40 and AGG<70.
16. To replace the column GRADE with 'C' in the table STUDENT whose AGG <40.
17. To display the TCODE and TEACHERNAME in ascending order of TCODE of all those teachers who teach the Humanities stream.
18. To display the NAME, STREAM and DATEOFJOINING from the table TEACHER in ascending order of STREAM and descending order of DATEOFJOINING within STREAM.
19. To display the highest BASIC within each STREAM from the table TEACHER

20. To display the grand total of the BASIC & average BASIC from the table TEACHER.
21. To display lowest AGG where TCODE is above 200 from the table STUDENT
22. To display the number of rows present in the table TEACHER
23. To display the average of AGG for each of the TCODE groupings from table STUDENT.
24. To find the total number of teachers in each STREAM.
25. To display the names of all the different STREAMs taught from the table TEACHER.
26. To increase the Basic by 1000, for all those teachers whose DATEOFJOINING is before 1st February 1995 ('1995-02-01').
27. To display the lowest BASIC within each STREAM in which 2 or more teachers are teaching.
28. To display NAME of the students and the corresponding TEACHERNAME, from the tables STUDENT and TEACHER with the help of corresponding entries for TCODE in both the tables.

IMPORTANT: REFER TO ASSIGNMENT BOOKLET FOR MORE QUESTIONS

General Instructions:

i. Type the solution of all the problems using SQL statements using **MySQL**

ii. Type in the following on top of each of your program as comment lines

```
# ------------------------------------------------------------#
# List-Program No     : L6-P1
# Developed By        : Eima Pathak
# Date                : 05-October-2025
# ------------------------------------------------------------#
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample output at the bottom of the program as comment lines.

iv. **"Turn in"** the softcopy of the programs as a single document in response to this assignment submission with each program on separate pages. Once verified by the teacher, take out a hardcopy from the printer and get them signed by the respective computer teacher. Add the printouts in the practical file and get the Index entry also signed.

v. Recommended Font, "**Courier New**", Style "**Bold**" Size "**11**" for all commands with single line spacing.

```
# ------------------------------------------------------------#
# List-Program No : L6-P1
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

CREATE TABLE TEACHER (
TCODE VARCHAR(3) PRIMARY KEY,
TEACHERNAME VARCHAR(25),
STREAM VARCHAR(15),
BASIC FLOAT,
DATEOFJOINING DATE
);

Query OK, 0 rows affected
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P2
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

INSERT INTO TEACHER VALUES
('101', 'ANANYA MURTY', 'SCIENCE', 18000, '1990-01-23'),
('202', 'KIRTI KHANEJA', 'HUMANITIES', 28000, '1978-12-12'),
('103', 'ADIL MEHRA', 'SCIENCE', 8000, '2001-02-14'),
('305', 'NISHIYA GOEL', 'COMMERCE', 12000, '1997-01-01'),
('203', 'SHUBH PANDIT', 'HUMANITIES', 22000, '1985-03-19'),
('109', 'NAINA', 'SCIENCE', 20000, '1994-07-17');

Query OK, 6 rows affected
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P3
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

CREATE TABLE STUDENT (
SCODE INT,
NAME VARCHAR(20),
TCODE VARCHAR(3),
AGG FLOAT
);

Query OK, 0 rows affected
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P4
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

INSERT INTO STUDENT VALUES
(2, 'NABI AHMAD', '101', 91),
(1, 'RAVI SAHAI', '305', 84),
(5, 'VIBHUTI', '203', 67),
(4, 'NAZNEEN', '103', 89),
(3, 'ARYANA', '202', 37),
(6, 'JONATHAN', '305', 45);

Query OK, 6 rows affected
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P5
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT * FROM TEACHER;
```

TCODE	TEACHERNAME	STREAM	BASIC	DATEOFJOINING
101	ANANYA MURTY	SCIENCE	18000	1990-01-23
103	ADIL MEHRA	SCIENCE	8000	2001-02-14
109	NAINA	SCIENCE	20000	1994-07-17
202	KIRTI KHANEJA	HUMANITIES	28000	1978-12-12
203	SHUBH PANDIT	HUMANITIES	22000	1985-03-19
305	NISHIYA GOEL	COMMERCE	12000	1997-01-01

```
# ------------------------------------------------------------#
# List-Program No : L6-P6
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT *
FROM TEACHER
WHERE DATEOFJOINING BETWEEN '1978-01-01' AND '1992-11-30';
```

TCODE	TEACHERNAME	STREAM	BASIC	DATEOFJOINING
101	ANANYA MURTY	SCIENCE	18000	1990-01-23
202	KIRTI KHANEJA	HUMANITIES	28000	1978-12-12
203	SHUBH PANDIT	HUMANITIES	22000	1985-03-19

```
# ------------------------------------------------------------#
# List-Program No : L6-P7
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT *
FROM STUDENT
ORDER BY AGG DESC;
```

SCODE	NAME	TCODE	AGG
2	NABI AHMAD	101	91
4	NAZNEEN	103	89
1	RAVI SAHAI	305	84
5	VIBHUTI	203	67
6	JONATHAN	305	45
3	ARYANA	202	37

```
# ------------------------------------------------------------#
# List-Program No : L6-P8
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT NAME, SCODE
FROM STUDENT
WHERE TCODE IN ('101', '203');
```

NAME	SCODE
NABI AHMAD	2
VIBHUTI	5

```
# ------------------------------------------------------------#
# List-Program No : L6-P9
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT TCODE, TEACHERNAME, BASIC, BASIC * 0.25 AS INCOMETAX
FROM TEACHER;
```

TCODE	TEACHERNAME	BASIC	INCOMETAX
101	ANANYA MURTY	18000	4500
103	ADIL MEHRA	8000	2000
109	NAINA	20000	5000
202	KIRTI KHANEJA	28000	7000
203	SHUBH PANDIT	22000	5500
305	NISHIYA GOEL	12000	3000

```
# ------------------------------------------------------------#
# List-Program No : L6-P10
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

UPDATE STUDENT
SET AGG = 82
WHERE NAME = 'RAVI SAHAI';

Query OK, 1 row affected
Rows matched: 1 Changed: 1 Warnings: 0
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P11
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

DELETE FROM STUDENT
WHERE AGG < 40;

Query OK, 1 row affected
Rows matched: 1 Changed: 1 Warnings: 0
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P12
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

UPDATE TEACHER
SET STREAM = 'HUMANITIES', TCODE = '210'
WHERE TEACHERNAME = 'ANANYA MURTY';

Query OK, 1 row affected
Rows matched: 1 Changed: 1 Warnings: 0
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P13
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

ALTER TABLE STUDENT
ADD GRADE CHAR(1);

Query OK, 0 rows affected
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P14
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

UPDATE STUDENT
SET GRADE = 'A'
WHERE AGG >= 70;

Query OK, 2 rows affected
Rows matched: 2 Changed: 2 Warnings: 0
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P15
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

UPDATE STUDENT
SET GRADE = 'B'
WHERE AGG >= 40 AND AGG < 70;

Query OK, 2 rows affected
Rows matched: 2 Changed: 2 Warnings: 0
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P16
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

UPDATE STUDENT
SET GRADE = 'C'
WHERE AGG < 40;

Query OK, 0 rows affected
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P17
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT TCODE, TEACHERNAME
FROM TEACHER
WHERE STREAM = 'HUMANITIES'
ORDER BY TCODE ASC;
```

TCODE	TEACHERNAME
202	KIRTI KHANEJA
203	SHUBH PANDIT
210	ANANYA MURTY

```
# ------------------------------------------------------------#
# List-Program No : L6-P18
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#
```

```
SELECT TEACHERNAME, STREAM, DATEOFJOINING
FROM TEACHER
ORDER BY STREAM ASC, DATEOFJOINING DESC;
```

TEACHERNAME	STREAM	DATEOFJOINING
NISHIYA GOEL	COMMERCE	1997-01-01
ANANYA MURTY	HUMANITIES	1990-01-23
SHUBH PANDIT	HUMANITIES	1985-03-19
KIRTI KHANEJA	HUMANITIES	1978-12-12
ADIL MEHRA	SCIENCE	2001-02-14
NAINA	SCIENCE	1994-07-17

```
# ------------------------------------------------------------#
# List-Program No : L6-P19
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT STREAM, MAX(BASIC) AS HIGHEST_BASIC
FROM TEACHER
GROUP BY STREAM;
```

STREAM	HIGHEST_BASIC
SCIENCE	20000
HUMANITIES	28000
COMMERCE	12000

```
# ------------------------------------------------------------#
# List-Program No : L6-P20
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT SUM(BASIC) AS GRAND_TOTAL_BASIC,
       AVG(BASIC) AS AVERAGE_BASIC
FROM TEACHER;
```

GRAND_TOTAL_BASIC	AVERAGE_BASIC
108000	18000

```
# ------------------------------------------------------------#
# List-Program No : L6-P21
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT MIN(AGG) AS LOWEST_AGG
FROM STUDENT
WHERE TCODE > 200;
```

LOWEST_AGG
45

```
# ------------------------------------------------------------#
# List-Program No : L6-P22
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT COUNT(*) AS TOTAL_ROWS
FROM TEACHER;
```

TOTAL_ROWS
6

```
# ------------------------------------------------------------#
# List-Program No : L6-P23
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT TCODE, AVG(AGG) AS Average_agg
FROM STUDENT
GROUP BY TCODE;
```

TCODE	Average_agg
101	91
305	63.5
203	67
103	89

```
# ------------------------------------------------------------#
# List-Program No : L6-P24
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT STREAM, COUNT(*) AS TOTAL_TEACHERS
FROM TEACHER
GROUP BY STREAM;
```

STREAM	TOTAL_TEACHERS
SCIENCE	2
HUMANITIES	3
COMMERCE	1

```
# ------------------------------------------------------------#
# List-Program No : L6-P25
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT DISTINCT STREAM
FROM TEACHER;
```

STREAM
SCIENCE
HUMANITIES
COMMERCE

```
# ------------------------------------------------------------#
# List-Program No : L6-P26
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

UPDATE TEACHER
SET BASIC = BASIC + 1000
WHERE DATEOFJOINING < '1995-02-01';
Rows matched: 4  Changed: 4  Warnings: 0
```

```
# ------------------------------------------------------------#
# List-Program No : L6-P27
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT STREAM, MIN(BASIC) AS LOWEST_BASIC
FROM TEACHER
GROUP BY STREAM
HAVING COUNT(*) >= 2;
```

STREAM	LOWEST_BASIC
SCIENCE	8000
HUMANITIES	19000

```
# ------------------------------------------------------------#
# List-Program No : L6-P28
# Developed By    : PUJA DEY
# Date            : 05-October-2025
# ------------------------------------------------------------#

SELECT STUDENT.NAME, TEACHER.TEACHERNAME
FROM STUDENT
INNER JOIN TEACHER
ON STUDENT.TCODE = TEACHER.TCODE;
```

NAME	TEACHERNAME
RAVI SAHAI	NISHIYA GOEL
VIBHUTI	SHUBH PANDIT
NAZNEEN	ADIL MEHRA
JONATHAN	NISHIYA GOEL